

Infrastructure Module Reference

Infrastructure Module Reference I

This section inventories every Layer-1 subdirectory returned by `22 discover_infrastructure_modules(repo_root)`. File totals use 456 Python sources across `infra` + 7,024 `infra` tests guarding them. Documentation Duality = paired `README.md` + `AGENTS.md`; optional `SKILL.md` manifests feed `python -m infrastructure.skills`.

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
<code>autoresearch</code>	7			<code>build_autoresearch_plan</code> , <code>readiness validation CLI</code>
<code>benchmark</code>	3			Template harness scoring + comparative gates
<code>config</code>	0			Repository defaults + hardened templates
<code>core</code>	97			<code>get_logger</code> , <code>load_config</code> , <code>TemplateError</code>
<code>docker</code>	0			Containerisation scaffolding
<code>doctor</code>	9			Checkout <code>diagnose/fix/undo repairs</code>
<code>documentation</code>	10			<code>FigureManager</code> , <code>generate_glossary</code>
<code>llm</code>	53			Ollama helpers, sanitization, review + translation pipelines

Infrastructure Module Reference II

<code>logrotate.d</code>	0	Rotation snippets (documentation-first)
<code>methods</code>	5	<code>build_methods_orchestration_plan</code> , <code>methods-stage</code> contracts + validation <code>PipelineRunner</code> , entry point for <code>./run.sh</code>
<code>orchestration</code>	8	<code>discover_projects</code> , <code>workspace</code> management
<code>project</code>	19	Markdown readability + prose tooling
<code>prose</code>	8	Zenodo, executable bundle, archival targets
<code>publishing</code>	42	BibTeX models, parsers, converters PDF/HTML/slide rendering, Pandoc filters
<code>reference</code>	8	Coverage parsers, dashboards, executive artefacts
<code>rendering</code>	45	<code>check_numerical_stability</code> , <code>benchmark_function</code>
<code>reporting</code>	51	<code>infrastructure.search.literature</code> clients + cache
<code>scientific</code>	6	<code>discover_skills</code> , SKILL manifest regeneration
<code>search</code>	8	
<code>skills</code>	6	

Infrastructure Module Reference III

steganography	10	Watermark overlays + hash manifests
validation	61	PDF + Markdown + integrity CLIs

Alphabetical summaries I

Below, `${module}_python_file_count` placeholders expand per subdirectory at render-time.

`infrastructure.autoresearch` (7 files)

Readiness planner, validation CLI, and report models for AutoResearch-style project promotion (`infrastructure/autoresearch/`).

`infrastructure.benchmark` (3 files)

Template harness scoring and comparative gate helpers exercised in CI smoke paths.

Repository-wide YAML templates and secure manifests (`.env.template`, hardened defaults referenced by Docker + CLI).

`infrastructure.core` (97 files)

Checkpointing, logging, pipeline YAML parsing, telemetry bridges, filesystem helpers, hardened exceptions. Everything else imports logging + error taxonomy from here first.

Alphabetical summaries II

`infrastructure.doctor` (9 files)

Checkout diagnose/fix/undo repairs for broken local workspace states.

`infrastructure.docker` (0 files)

Pinned images / compose scaffolding for reproducible CI + remote builds.

`infrastructure.documentation` (10 files)

Figure registries plus glossary tooling feeding manuscript automation.

`infrastructure.llm` (53 files)

Ollama integrations, sanitization adapters, templated reviewer flows. **Literature ingestion now lives primarily in `search/literature` + citation helpers in `reference/`.**

Alphabetical summaries III

`infrastructure.orchestration` (8 files)

`python -m infrastructure.orchestration` exposes interactive menus, subprocess wiring for thin shell wrappers (`run.sh`, `secure_run.sh`), and stubs used in CI for menu parsing tests.

`infrastructure.project` (19 files)

Canonical discovery (`discover_projects`) enforcing `src/` + `tests/`, slug validation, nested WIP namespaces.

`infrastructure.prose` (8 files)

Readability metrics + Markdown tooling for prose-centric manuscripts / CI gates.

`infrastructure.publishing` (42 files)

Metadata models, APA/BibTeX/MLA formatters, optional Zenodo clients.

Alphabetical summaries IV

`infrastructure.reference` (8 files)

Citation/BibTeX parsing + conversion utilities leveraged by manuscripts and retrieval scripts.

`infrastructure.rendering` (45 files)

Pandoc shim, Unicode/XeLaTeX postprocessors, combined PDF/HTML/slide exporters.

`infrastructure.reporting` (51 files)

Parses pytest + coverage artefacts for dashboards; pairs with Stage 01 summaries and downstream executive exports.

`infrastructure.scientific` (6 files)

Stability probing, benchmarking hooks—consumed heavily by optimization exemplars (`template_code_project` scripts).

Alphabetical summaries V

`infrastructure.search` (8 files)

literature/ client stack (client.py, backends, caches)
powering archive-only template_search_project literature
workflows when copied locally from projects_archive/.

`infrastructure.skills` (6 files)

Discovers SKILL.md frontmatter → .cursor/skill_manifest.json.

`infrastructure.steganography` (10 files)

Watermark overlays, hashing companions triggered by secure pipeline path.

`infrastructure.validation` (61 files)

Markdown + PDF + integrity CLIs underpinning Stage 04 diagnostics.

Alphabetical summaries VI

[infrastructure/logrotate.d](#) (0 files)

Operational templates for deployments (documentation-first; intentionally minimal Python footprint).

Documentation maturity: Coverage statements in Results pull from introspection—not hand-maintained denominators—so newly promoted modules automatically flow into manuscripts after `generate_manuscript_metrics.py`.

FAIR+RSE linkage: MCP-ready SKILL.md artefacts align with evaluator heuristics (executability + metadata richness) emphasized by FAIRsoft guidance [[@garijo2024fairsoft](#)].